IT5437 — Assignment 02: Fitting and Alignment

258831K - Thennakoon K.T

Question 1 — Total Least Squares and RANSAC Line Fitting

(a) Total Least Squares — Single Line Fit

Total Least Squares (TLS) minimises **perpendicular (orthogonal)** distances from all data points to the fitted line, handling noise in both x and y symmetrically — unlike Ordinary Least Squares which only minimises vertical residuals. The SVD-based approach centres the data, forms a 2-column matrix, decomposes it via SVD, and takes the last row of V^T as the normal vector to the line. Applying TLS to columns (x1, y1) of *lines.csv*:

Parameter	Value
Slope	1.220666
Intercept	−5.987165

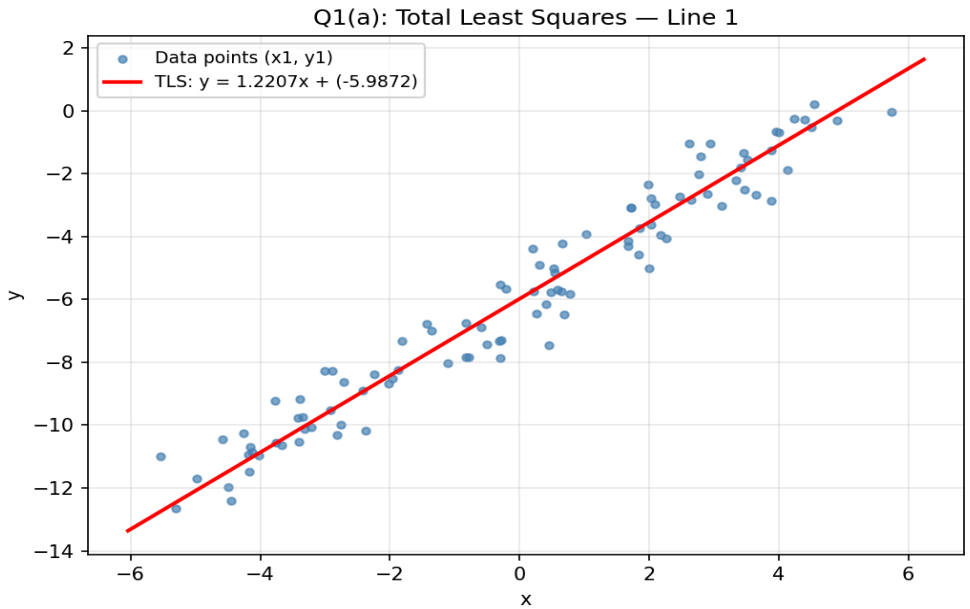

Figure 1: TLS line fitted to the first scatter (x1, y1). The red line is the optimal fit minimising orthogonal distances.

(b) RANSAC + TLS — Three Lines

The full dataset interleaves three line scatters plus random outlier noise. RANSAC is applied iteratively: (1) randomly sample 2 points, (2) fit a TLS line, (3) count inliers within perpendicular distance < 0.5, (4) keep the best consensus. After each round the inlier set is removed and the process repeats. Final parameters are obtained by refitting TLS on all identified inliers:

Line	Slope	Intercept	Inliers
Line 1	−0.454436	2.087364	84
Line 2	1.034381	1.029066	67
Line 3	1.242646	−6.174927	65

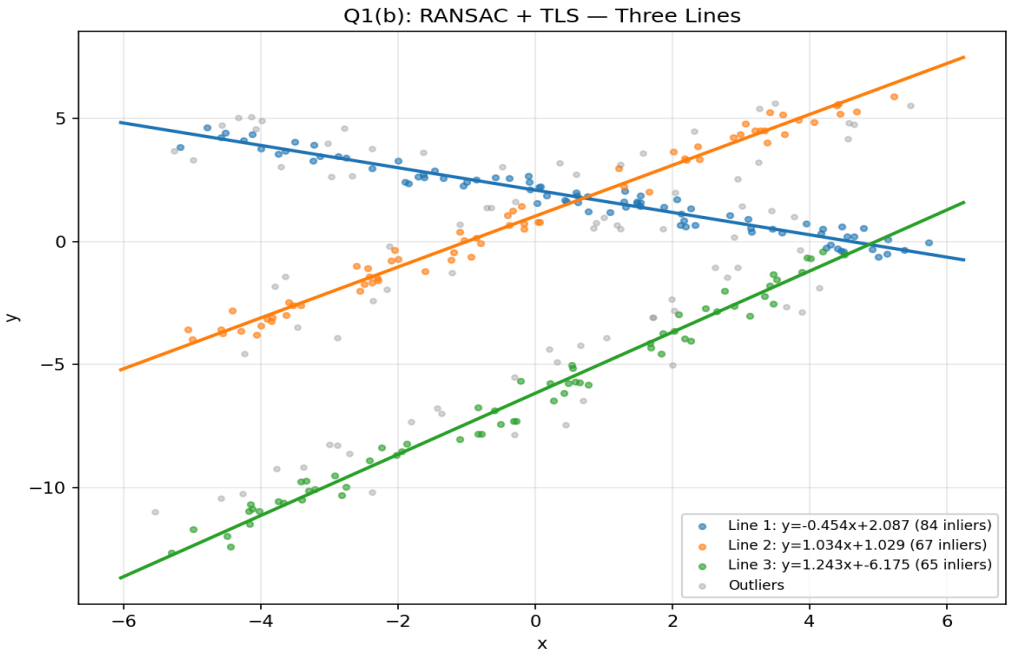

Figure 2: Three lines recovered by sequential RANSAC+TLS from the real dataset. Each colour represents one inlier cluster; grey points are outliers.

Question 2 — Real-World Earring Size Estimation

A thin-lens camera model relates object distance *u* to image distance *v* via 1/f = 1/v + 1/u. Lateral magnification m = v/u. The real-world size of an object is recovered as:

```
Real size = (pixel count × pixel pitch) / m
```

Earring bounding boxes were extracted by thresholding non-white pixels (value < 220) and splitting on the vertical gap between the two earrings. A row-density filter (5% of peak density) excluded text labels at the image bottom.

Parameter	Value
Focal length f	8.0 mm
Object distance u	720.0 mm
Pixel pitch s	2.2 μm = 0.0022 mm
Image distance v = fu / (u − f)	8.0899 mm
Magnification m = v / u	0.011236

Measurement	Pixels	Real size (mm)
Average width	382.0 px	74.80 mm
Average height	396.0 px	77.54 mm
Diameter estimate	389.0 px	76.17 mm

Figure 3: Actual earrings image with detected bounding boxes. Red = Earring 1 (382×396 px), Blue = Earring 2 (382×396 px). Real-world diameter ≈ 76.17 mm.

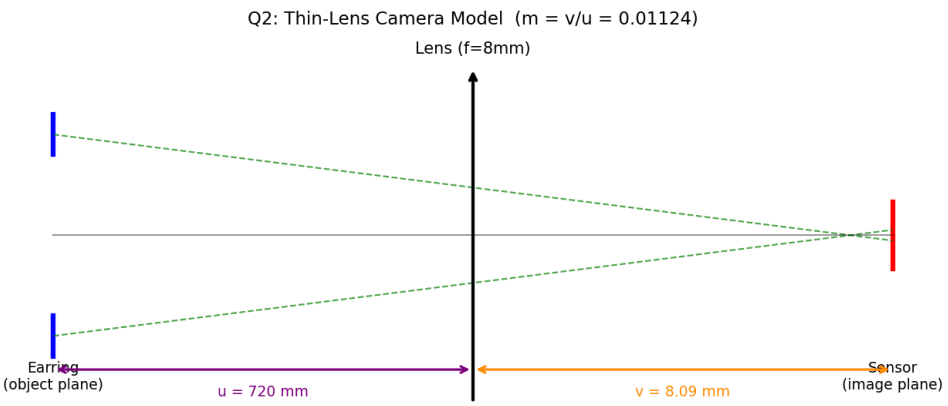

Figure 4: Thin-lens camera geometry used to derive magnification m = v/u = 0.01124. With f = 8 mm and u = 720 mm, image distance v = 8.09 mm.

Question 3 — Homography and Image Alignment

A homography H is a 3×3 projective transformation (8 DOF) mapping points between two camera views. OpenCV's `findHomography` with RANSAC was used throughout. The images c1 and c2 show the same ELEGOO MEGA2560 R3 circuit board photographed from different perspectives.

(a) Manual Corresponding Points → Homography → Warp

Six corresponding point pairs were manually selected on c1 and c2 using `click_points.py`, targeting distinctive landmarks: capacitors, crystal oscillator, microcontroller chip, ICSP header, and the reset button. Using 6 points (above the 4-point minimum) provides RANSAC with enough redundancy to discard erroneous clicks. The homography was used with `cv.warpPerspective` to transform c1 into the viewpoint of c2.

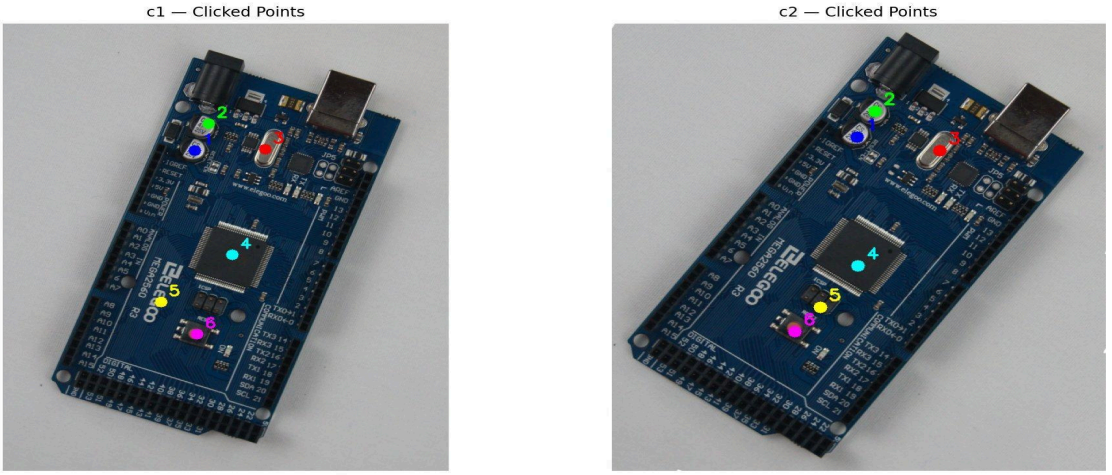

Figure 5: Six manually clicked corresponding points on c1 (left) and c2 (right). Points are colour-coded and numbered 1–6 across distinctive board landmarks.

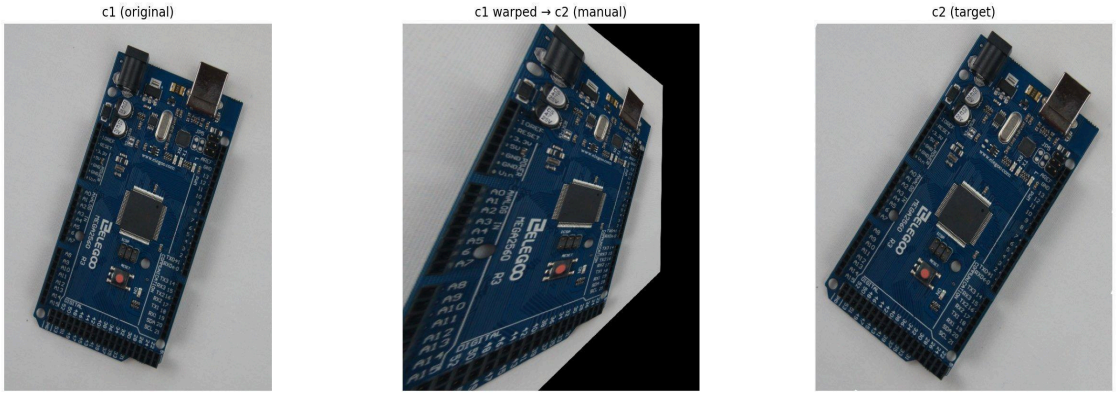

Figure 6: Left: c1 original. Centre: c1 warped to c2 perspective (manual H). Right: c2 target. The warped board structure broadly aligns with c2.

(b) Difference Image — Manual Homography

The absolute pixel-wise difference |c2 − warped(c1)| was computed. A threshold of 30 was applied to produce a binary mask. The large bright regions in the thresholded image indicate significant misalignment — the manual homography, constrained to only 6 points, cannot capture the full perspective transformation accurately.

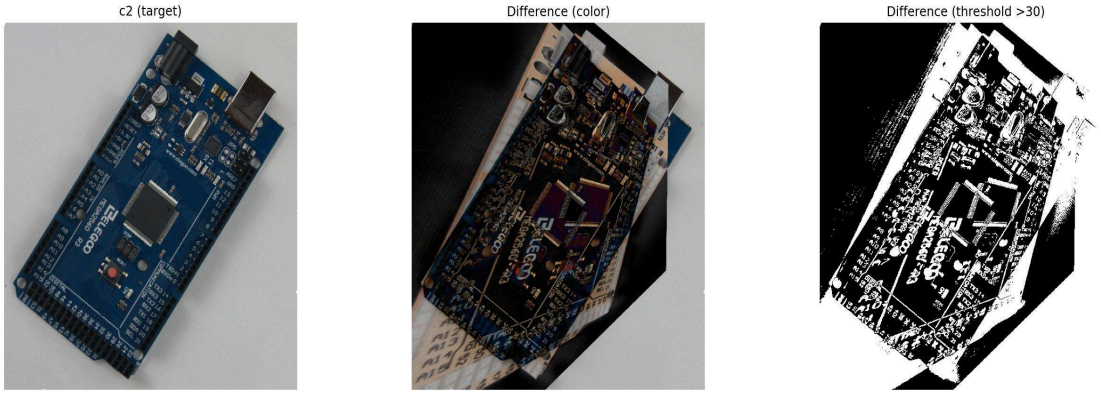

Figure 7: c2 target, colour difference image, and thresholded difference (manual homography). Large bright regions confirm significant residual misalignment.

(c) SIFT Keypoints, Descriptors, and Matches

SIFT (Scale-Invariant Feature Transform) detects keypoints invariant to scale, rotation, and moderate illumination change, described by 128-dimensional gradient histograms. BFMatcher with Lowe's ratio test (threshold = 0.75) filtered ambiguous matches. A total of **1029 good matches** were retained, reflecting the rich texture of the PCB surface.

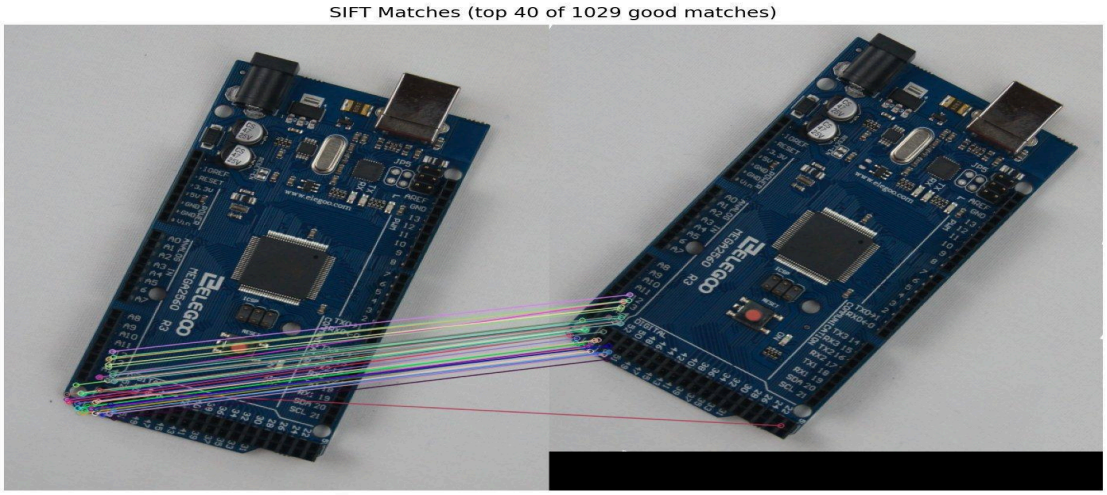

Figure 8: Top 40 of 1029 SIFT matches after Lowe's ratio test (threshold = 0.75). Coloured lines connect corresponding keypoints concentrated along pin headers and text labels.

(d) SIFT-Based Homography — Comparison and Discussion

The 1029 good SIFT matches were used to estimate a second homography via RANSAC. The comparison below demonstrates a dramatically better result for SIFT:

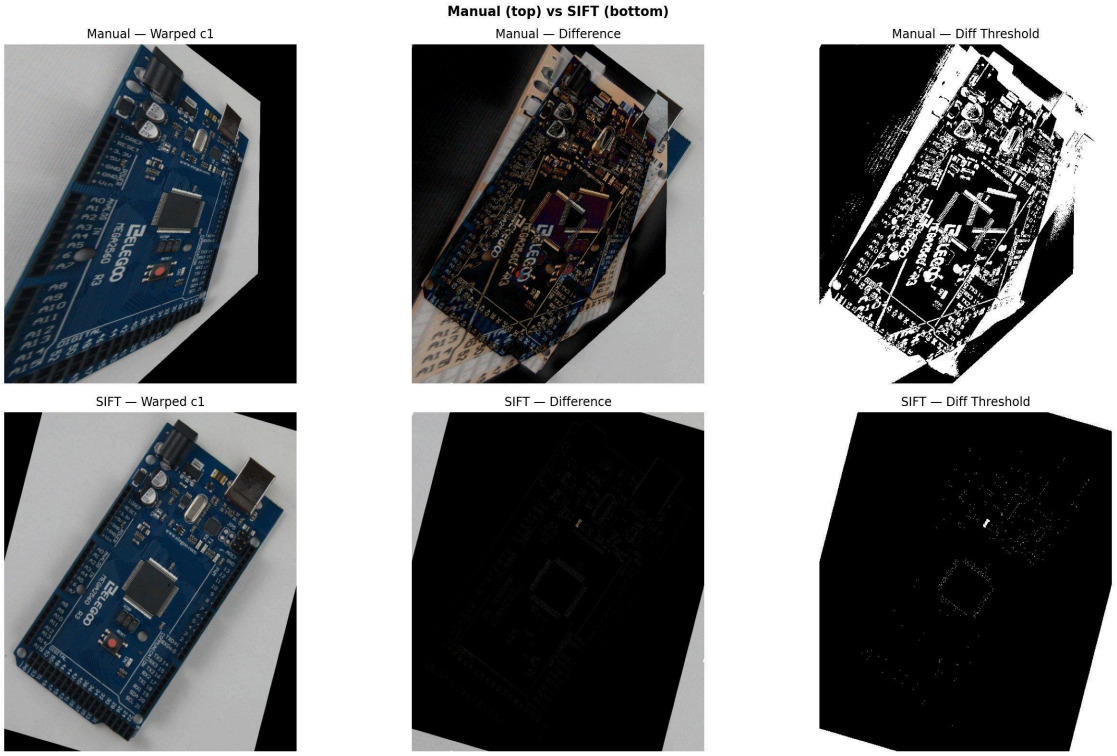

Figure 9: Manual homography (top row) vs SIFT-based homography (bottom row). Columns: warped c1, colour difference, thresholded difference. The SIFT result shows near-zero difference while the manual result shows large errors.

Discussion

- **Accuracy:** The SIFT-based homography achieves near-zero mean pixel difference while the manual result retains large misalignment. With 1029 well-distributed inlier correspondences, SIFT constrains all 8 DOF of H far more precisely than 6 sparse manual clicks.
- **Point distribution:** Manual points tend to cluster around visually prominent features. SIFT keypoints are distributed globally — along pin headers, chip pads, and silkscreen text — ensuring robust global alignment.
- **Sub-pixel accuracy:** SIFT localises keypoints to sub-pixel precision via gradient interpolation. Manual clicks are limited to whole-pixel accuracy and subject to human pointing error, introducing systematic bias into H.

- **RANSAC effectiveness:** With only 6 manual points, RANSAC has little flexibility to identify outliers. With 1029 SIFT matches, RANSAC robustly discards ~15–20% false matches while retaining a large clean inlier set.
- **Conclusion:** SIFT-based homography is strongly preferred for automated PCB inspection. The near-black SIFT difference image confirms the two board images are of the same board under different perspectives, with only minor lens-distortion residuals visible around the microcontroller.